

Factor Timing

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The optimal factor timing portfolio is equivalent to the stochastic discount factor. We propose and implement a method to characterize both empirically. Our approach imposes restrictions on the dynamics of expected returns, leading to an economically plausible SDF. Market-neutral equity factors are strongly and robustly predictable. Exploiting this predictability leads to substantial improvement in portfolio performance relative to static factor investing. The variance of the corresponding SDF is larger, is more variable over time, and exhibits different cyclical behavior than estimates ignoring this fact. These results pose new challenges for theories that aim to match the cross-section of stock returns. (*JEL* G12, G14, G17)

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Aggregate stock returns are predictable over time (e.g., Shiller 1981; Fama and French 1988), creating the scope for investors to engage in market timing. Factors beyond the aggregate market are sources of risk premiums in the cross-section of assets (e.g., Fama and French 1993), creating the basis for factor investing. How valuable is it to combine these two ideas and construct an optimal *factor timing* portfolio that unifies cross-sectional and time-series predictability of returns? Answering this question has economic importance:

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the optimal portfolio is equivalent to the stochastic discount factor (SDF). Therefore, if factor timing is relevant for the optimal portfolio, we should account for this fact when estimating the SDF.

Empirically determining the value of factor timing appears difficult, because doing so requires measuring the predictability of many returns and opens the door for spurious findings. We propose a new approach to overcome this challenge. Imposing that the implied SDF is not too volatile leads us to focus on estimation of predictability of the largest principal components of the factors. We find that these statistical restrictions are crucial to construct robust forecasts of factor returns.

Taking into account the predictability of the factors leads to an estimated SDF, which exhibits drastically different properties than estimates that assume constant factor premiums, the standard approach in previous work. Our estimated SDF is more volatile: its variance increases from 1.66 to 2.96. Moreover, that the benefits to factor timing are strongly time varying results in much more heteroscedasticity of the SDF. These fluctuations in SDF variance exhibit a very different pattern than estimates that only account for the predictability of market returns. They mostly occur at shorter business-cycle frequencies, and are correlated with different macroeconomic variables.

Our empirical analysis focuses on fifty standard stock “anomaly” portfolios that have been put forward in previous work as capturing cross-sectional variation in expected returns. To characterize an optimal portfolio and the SDF, we rely on two restrictions. First, we assume that the SDF that prices stocks implies that a *conditional* factor model holds with respect to these portfolios. Our setting enriches the previous literature by allowing for time-varying loadings of the SDF on the factors. With this assumption, we entertain the possibility that factor timing strategies are profitable. Second, we assume that the pricing kernel does not generate excessively high Sharpe ratios.¹ Such near-arbitrage opportunities would be eliminated by arbitrageurs in equilibrium (Kozak et al. 2018). When the cross-section of returns has a strong factor structure, this assumption implies that the time-series variation in risk premiums is mostly driven by time-varying exposure of the SDF to the largest sources of variation in realized returns. If this were not the case, small components would be highly predictable, generating implausibly large Sharpe ratios. The fifty portfolios we consider exhibit a factor structure with a stable covariance matrix over time that allows us to exploit this idea empirically. We focus on the largest sources of variation by restricting our attention to the first five principal components (PCs) of anomaly returns, because these components explain 60% of the variation in realized returns. This dimension reduction allows for robust estimation of their predictability, and therefore the SDF. As such, our approach is a regularization of the left-hand side of the predictability problem —“which

¹ Using simple economically motivated restrictions on asset prices to stabilize statistical inference has notable antecedents (see, e.g., Cochrane and Saa-Requejo 2000; Campbell and Thompson 2007).

factors are predictable?”—rather than the right-hand side—“which variables are useful predictors?” We take a simple stance on this second issue by using only the book-to-market ratio of each portfolio as a measure to predict its returns.

We find that the PCs of anomalies are strongly predictable. For the two most predictable components, the first and fourth PCs, their own book-to-market ratios predict future monthly returns with an out-of-sample R^2 around 4%, about 4 times larger than that of predicting the aggregate market return. We confirm these strong relations are not driven by statistical issues arising in small samples. The predictability of the dominant PC portfolios captures common variation in risk premiums that allows us to form forecasts of individual anomaly returns. These forecasts yield a sizable total out-of-sample monthly R^2 around 1%. The observation that factor returns are robustly predictable lends support to the enterprise of factor timing. We confirm that this conclusion does not rely on the details of our implementation by varying, for example, the number of principal components, the construction of anomaly portfolios, and the horizon of predictability.

The key ingredient for our results is the dimension reduction of the set of portfolios to predict. One could instead separately estimate expected returns for each anomaly portfolio without recognizing the factor structure in returns and imposing the absence of near arbitrage. We find this approach to be less fruitful: predicting each anomaly return with its own book-to-market ratio generates only half of the predictability our restrictions uncover. The out-of-sample robustness of our approach supports the validity of these restrictions. Interestingly, our approach complements the set of methods developed to choose among many predictors given a portfolio to predict. For example, we find that the 3-pass regression filter of Kelly and Pruitt (2013a) provides more robust results when applied to the dominant components of anomalies rather than to each portfolio separately.

We use our results to construct an optimal factor timing portfolio. This allows us to quantify the investment benefits to factor timing. And, more importantly, we use it to characterize the properties of an SDF consistent with the evidence of these factor timing benefits. First, timing expected returns provides substantial investment gains; a pure factor timing portfolio achieves a Sharpe ratio of 0.71. This means that the conditional variance of the SDF is substantially larger than that inferred from static strategies alone. The benefits from timing market-neutral factors largely outweigh those from timing the aggregate market return and are comparable to those obtained by static factor investing. Second, these benefits vary over time: the SDF is strongly heteroscedastic. Variation in the maximum compensation for risk is driven by changes in the means of the factors and to a lesser extent changes in their variances. Again, these fluctuations are much more pronounced than fluctuations in the Sharpe ratio of the market portfolio. Third, the dynamics of the variance of the SDF differ from those of the market risk premium. The SDF variance evolves mostly at business cycle

frequency rather than at longer horizons. However, it is not always related to recessions. More broadly, macroeconomic variables capturing variations in the price of market risk often have different relations with the SDF variance. Fourth, the contribution of various anomalies to the SDF exhibit interesting dynamics. For example, the loadings of size and value are procyclical while the loading of momentum is countercyclical.

To summarize, factor timing is very valuable, above and beyond market timing and factor investing taken separately. The changing conditional properties of the pricing kernel are mostly driven by market-neutral factors. The methods and facts we study in this paper are only the beginning of the economic enterprise of understanding the evolution of drivers of risk premiums. Our results suggest that theories developed to understand cyclical variation in the price of market risk (e.g., Bansal and Yaron 2004; Barberis et al. 2015; Campbell and Cochrane 1999; Campbell and Kyle 1993) are unlikely to capture the *dynamic* properties of the cross-section of returns. Indeed, these models generate SDFs that are much less volatile and heteroscedastic than our estimated SDF. Further, they typically focus on a single common force driving variation in risk premiums, at odds with the multiple dimensions we uncover. Finally, the properties of our estimated SDF provide a useful set of moments summarizing the properties of the rich cross-section of stocks, moments that future theories should target.

This paper builds on a long literature that studies the time-series predictability of returns starting from Shiller (1981) and Fama and French (1988) for stocks and Fama and Bliss (1987) for bonds.² Although this early evidence is mostly about aggregate returns, our main focus is to understand predictability of the cross-sections of returns. Early work has extended the ideas of market predictability to specific anomalies: Cohen, Polk, and Vuolteenaho (2003) for value or Cooper, Gutierrez, and Hameed (2004) and Daniel and Moskowitz (2016) for momentum. We aim to tackle the entire cross-section. With a similar goal, various papers, such as those of Stambaugh, Yu, and Yuan (2012) and Akbas et al. (2015), examine the ability of a single variable to forecast all anomaly returns and thus their common component. **These papers implicitly assume a single source of time-varying risk premiums;** our approach entertains multiple. This multiplicity is complementary to that of Kelly and Pruitt (2013a): we study how to predict many returns with a factor-specific explanatory variable (its own valuation ratio), whereas they predict a single return with a wide cross-section of valuation ratios. That is, we ask a question of what we should predict, whereas they study how to select variables that are most useful in making such a prediction.

Another strand of the literature studies the predictability of returns anomaly by anomaly (or stock by stock) without imposing any structure on the

² See Kojien and Van Nieuwerburgh (2011) for a survey of recent work on the topic.

implied pricing kernel. Recent prominent examples are Campbell, Polk, and Vuolteenaho (2009) and Lochstoer and Tetlock (2020), **who use panel VAR techniques to forecast firm-level expected returns, then aggregate the estimates into portfolios**. Arnott, Beck, and Kalesnik (2016a,b), Asness et al. (2000), Baba Yara et al. (2018), Cohen, Polk, and Vuolteenaho (2003), and others use **valuation ratios to forecast anomaly returns**. Greenwood and Hanson (2012) forecast characteristics-based anomalies using their “issuer-purchaser” spread—the difference in the average characteristic for net equity issuers versus repurchasers. Conversely, some papers, such as those by Ilmanen, Nielsen, and Chandra (2015), Asness (2016), and Asness et al. (2017), find that cross-sectional long-short factors are not very predictable by valuation ratios. Regardless of their conclusion, all of these papers forecast a single return at a time, ignoring the correlation across assets. Implicitly, they assume there are potentially as many independent sources of time-varying risk premiums as there are assets. **We, instead, study common sources of predictability across all anomalies in a restricted setup and then infer the implied predictability of each anomaly**. Such an approach brings important statistical advantages in terms of dimensionality reduction. In Section 2, we show our method yields greater out-of-sample predictability than various alternatives.

Another literature develops methods for dealing with the large dimensionality of the cross-section. Freyberger, Neuhierl, and Weber (2018) use an adaptive group lasso method to test which characteristics provide independent information for the cross-section of expected returns on individual stocks. Kozak et al. (2020) model SDF risk prices as linear functions of characteristics, while Kozak (2019) extends this approach to capture arbitrary nonlinearities. Kelly, Pruitt, and Su (2019) model stock betas as linear functions of characteristics. Light, Maslov, and Rytchkov (2017), Kelly, Pruitt, and Su (2019), and Giglio and Xiu (2018) employ latent factor analysis. Kozak et al. (2018, 2020) use no near-arbitrage to argue for the use of principal components analysis (PCA). All these dimension-reduction techniques are somewhat related. For example, Kelly, Pruitt, and Su (2019) show that if the cross-sectional correlation matrix of stock characteristics is constant their latent factors exactly correspond to largest PCs of characteristics-managed anomaly portfolios. We also use PCA to handle a high-dimensional factor space. However, we differ from this previous work in an important dimension. In all these papers, expected returns on anomaly portfolios are approximately constant or vary only due to time-varying volatility. In contrast, we entertain the possibility and find evidence for significant time variation in prices of risk on these factors.

Finally, some papers highlight the quantitative importance of conditioning information for the SDF. Gallant, Hansen, and Tauchen (1990) use conditioning information and asset prices to derive lower bounds on the unconditional variance of the SDF. We use upper bounds on the unconditional variance to derive restrictions on the impact of conditioning information.

Chernov, Lochstoer, and Lundebj (2018) propose testing asset-pricing models using multihorizon returns and find that many standard empirical models of the SDF are rejected, exactly because they lack time variation in how the SDF loads on the factors. Moreira and Muir (2017) document benefits to volatility timing, implying changes in volatility play a meaningful role in the heteroscedasticity of the SDF.

1. Methodology

We are interested in assessing the benefits of timing strategies for a cross-section of excess returns $\{R_{i,t}\}_{i \in I}$; our main empirical setting is the cross-section of stock returns. Studying these timing benefits is important not only for the purpose of optimal portfolio choice but also to understand the economic forces shaping equilibrium prices. Calculating these benefits requires measuring the dynamics of risk premiums. The connection between factor timing benefits, the stochastic discount factor, and predictability is best illustrated by the following decomposition. In Internet Appendix Section I, we show that **if asset returns are uncorrelated**, the average maximum conditional Sharpe ratio can be expressed as

$$\mathbb{E}(\text{SR}_i^2) = \mathbb{E}[\text{var}_t(m_{t+1})] = \sum_i \frac{\mathbb{E}[R_{i,t+1}]^2}{\sigma_i^2} + \sum_i \left(\frac{R_i^2}{1 - R_i^2} \right). \quad (1)$$

The first equality shows that the average maximum squared Sharpe ratio coincides with the expected variance of m_{t+1} , where **m_{t+1} is the minimum variance stochastic discount factor that prices the set of returns**. The second equality shows that these quantities combine two elements. The first term is an unconditional part reminiscent of static Sharpe ratios: the sum of ratios of the squared average return to σ_i^2 , **the conditional variance of the asset return**.³ Of interest to us is the second term, which encodes predictability. This term is increasing in the R_i^2 s, **the maximum predictive R -squared when forecasting asset i** , $R_i^2 = 1 - \sigma_i^2 / \text{var}(r_{i,t+1})$.

Without any structure, it is challenging to create robust forecasts for all returns. Spurious results are likely, especially with many assets to predict. In this section, we show how two simple restrictions help address these issues. **First, we follow the literature and assume that a relatively small number of stock characteristics capture pricing-relevant information**. Equivalently, the assets are conditionally priced by a factor model, the main motivation behind factor timing portfolio strategies. **Second, we assume that prices feature no near-arbitrage opportunities**. **These assumptions imply that measuring the predictability of the largest principal components of the set of factors is enough to characterize**

³ For simplicity of exposition, we focus on a homoscedastic setting.

expected returns.⁴ This strong dimension reduction allows us to use the standard tools for forecasting single return series to measure this predictability.

1.1 Factor model, factor investing, and factor timing

First, we impose some structure on the pricing kernel. Start with the minimum variance SDF in the span of N individual stock (asset) excess returns R_{t+1} (Hansen and Jagannathan 1991):

$$m_{t+1} = 1 - b'_t(R_{t+1} - \mathbb{E}_t[R_{t+1}]), \quad (2)$$

which satisfies the fundamental relation $0 = \mathbb{E}_t[m_{t+1}R_{t+1}]$. We restrict the behavior of the loading b_t . As in Kelly, Pruitt, and Su (2019), Kozak et al. (2020), Freyberger, Neuhierl, and Weber (2018), and Kozak (2019), we use stock characteristics to reduce the dimensionality of the return space and the SDF. In particular, we assume that cross-sectional heterogeneity in risk prices b_t can be largely captured by K observable stock characteristics, C_t , with $K \ll N$. Time-series variation in the importance of each characteristic is summarized by the vector δ_t of size $K \times 1$.

Assumption 1. Stock-level SDF loadings can be represented as

$$\underbrace{b_t}_{N \times 1} = \underbrace{C_t}_{N \times K} \underbrace{\delta_t}_{K \times 1}, \quad (3)$$

where C_t is an $N \times K$ matrix of stock characteristics and δ_t is a $K \times 1$ vector of (possibly) time-varying coefficients, and $K \ll N$.

Substituting Equation (3) into Equation (2), we obtain an alternative SDF representation

$$m_{t+1} = 1 - \delta'_t(F_{t+1} - \mathbb{E}_t[F_{t+1}]), \quad (4)$$

where $F_t = C'_{t-1}R_t$ are “characteristics-managed” factor portfolios. For example, if one element of C_t is the market capitalization of a firm, the corresponding factor is the market return. If the characteristic is an indicator taking values -1, 0, or 1 depending on whether a stock is in the upper or lower quantiles of a characteristic, the corresponding factor is a standard sort-based portfolio in the style of Fama and French (1992). We can now interpret δ_t as time-varying prices of risk on these factor portfolios.

Therefore, Assumption 1 allows us to use characteristics to not only parsimoniously describe the cross-section but also to permit the time variation in a relatively small number of factor risk prices. Consequently, it results in a large dimensionality reduction in the number of variables determining the

⁴ Specifically, the expected return on any asset is the product of the asset’s conditional loading on and the expected return of these few components.

SDF. However, it is rich enough to consider meaningful variation in factor expected returns, the fundamental idea behind factor timing. For example, this is a richer setting than in Kozak et al. (2020), who assumes the mapping from stock characteristics to SDF coefficients is constant, $b_t = C_t \delta$.⁵ In such a specification, there is no scope for factor timing, because SDF coefficients are equal to weights in the maximum Sharpe ratio portfolio. Another way to see that our model entertains factor timing is to notice that our SDF can be alternatively represented as a factor model, with arbitrarily changing expected factor returns.

Lemma 1. (Conditional factor model) A conditional factor model holds:

$$\mathbb{E}_t [R_{j,t+1}] = \beta'_{jt} \Sigma_{F,t} \delta_t = \beta'_{jt} \mathbb{E}_t [F_{t+1}]. \quad (5)$$

The equivalence between the factor model and Equation (4) is given by $\delta'_t = \Sigma_{F,t}^{-1} \mathbb{E}_t (F_{t+1})$, where $\Sigma_{F,t}$ is the conditional covariance matrix of the factors. This relation also highlights that our model can generate interesting risk premium variation even in a homoscedastic setting for returns, where the variance-covariance matrix of the factors $\Sigma_{F,t}$ and the betas β'_{jt} are constant over time. This result arises because δ_t controls how the SDF loads on the factors, and therefore changes their price of risk.⁶

In our framework, because the factors completely capture the sources of risks of concern to investors, optimal portfolios can be constructed from only these few factors, the so-called “mutual fund theorem.” Factor timing strategies are the dynamic counterpart of this observation; as the properties of the factors change, an investor should adjust her portfolio weights accordingly. For example, the maximum conditional Sharpe ratio return is obtained by

$$R_{t+1}^{\text{opt}} = \mathbb{E}_t [F_{t+1}]' \Sigma_{F,t}^{-1} F_{t+1}. \quad (6)$$

Knowledge of the conditional risk premiums of the factors is crucial to form this and other dynamic strategies.

While going from individual assets to factors provides some useful dimension reduction and stabilization of the covariance structure, we are still left with many factor returns to forecast.⁷ A multitude of empirical results and theoretical motivations have put forward a large number of potential factors, leading to the

⁵ Although they focus on slightly different functional forms, Kelly, Pruitt, and Su (2019) and Freyberger, Neuhierl, and Weber (2018) also assume a constant mapping between characteristics and factor exposures.

⁶ Variations in δ_t could occur for multiple reasons. For example, investors' aversion to the various sources of factor risk could change over time. Or their exposure to these risks, for example, through consumption, could change over time.

⁷ This stabilization role is discussed in, for example, Brandt, Santa-Clara, and Valkanov (2009), Cochrane (2011), and Kozak et al. (2018).

emergence of what Cochrane (2011) calls the “factor zoo.” Including potentially irrelevant factors does not affect the theoretical performance of a factor model; the SDF would just have zero loading on these factors. However, including too many factors leads to greater probability of estimating spurious return predictability in finite samples. We now turn to a second assumption, which helps discipline our empirical analysis.

1.2 Absence of near-arbitrage

Various authors have used the idea of the absence of “good deals” or “no near-arbitrage” opportunities to add economic discipline to statistical exercises. For example, Cochrane and Saa-Requejo (2000) impose an upper bound on the conditional variance of the SDF to derive bounds on option prices. Ross (1976) originally proposed a bound on the squared Sharpe ratio for an unconditional empirical implementation of his APT in a finite-asset economy. Such a bound on the maximum squared Sharpe ratio is immediately equivalent to an upper bound on the variance of the SDF, m_{t+1} (Hansen and Jagannathan 1991). Kozak et al. (2018) use a similar argument to show that unconditionally, the large principal components of anomaly returns must explain most of the cross-sectional variation in average returns. In our setting with time-varying risk premiums, there is no such thing as *the* maximum Sharpe ratio, rather a conditional maximum Sharpe ratio at each point in time. What is an appropriate metric for a “good deal in this setting?” We argue that it is the average maximum conditional squared Sharpe ratio, $\mathbb{E}[\text{SR}_t^2]$.

Assumption 2. (Absence of near-arbitrage) There are usually no near-arbitrage opportunities: average conditional squared Sharpe ratios are bounded above by a constant.

Two interpretations of this restriction on Sharpe ratios help clarify the restriction’s economic content. First, this quantity corresponds to the average certainty equivalent for a mean-variance investor who optimally uses conditioning information. For such an investor with risk-aversion parameter γ , her certainty equivalent at time t is $\frac{\text{SR}_t^2}{2\gamma}$ where SR_t^2 is the maximum conditional squared Sharpe ratio (see Internet Appendix Section I). Taking the unconditional expectation of SR_t^2 measures, on average, the welfare gain from investing in risky assets. Second, the average conditional variance of the SDF in Equation (4) is also equal to its unconditional variance, because it has constant conditional mean. Therefore, our bound is also a bound on the unconditional variance of the SDF. Third, it equivalently provides an upper bound on the maximum unconditional squared Sharpe ratio when considering all possible dynamic factor strategies. This unconditional value measures the welfare gain for a mean-variance investor who does not have direct access to conditioning information but follows a buy and hold strategy from a menu of

managed portfolios (Ferson and Siegel 2001).⁸ Both of these interpretations highlight that our bound forbids “good deals” on average, but not always.

We now show that Assumption 2 leads to further dimensionality reduction. First, notice that because the maximum conditional Sharpe ratio is invariant to rotations of the asset space, we can apply Equation (1) with the PC decomposition of returns. Letting $PC_{i,t+1}$ be the i th principal component portfolios of the factors F , and λ_i the corresponding eigenvalue, we have

$$\mathbb{E}(\text{SR}_t^2) = \sum_{i=1}^K \frac{\mathbb{E}[PC_{i,t+1}]^2}{\lambda_i} + \sum_{i=1}^K \left(\frac{R_i^2}{1 - R_i^2} \right), \quad (7)$$

where the summation is across all K PC portfolios. Again, the first term represents the benefits of static factor investing. It is the squared Sharpe ratio of an optimal static factor portfolio. The second term is our focus in this paper and captures the amount of predictability for each principal component. The more a principal component can be predicted, the better portfolio performance an investor can obtain. This second term represents the incremental benefit of optimally timing the factors.

Second, we can ask how much each PC portfolio contributes to the total predictability of returns. As a measure of the total amount of predictability we define the total R^2 as

$$\begin{aligned} R_{\text{total}}^2 &\equiv \frac{\text{tr}[\text{cov}(\mathbb{E}_t[F_{t+1}])]}{\text{tr}[\text{cov}(F_{t+1})]} = \frac{\text{tr}[\text{cov}(\mathbb{E}_t[PC_{t+1}])]}{\text{tr}[\text{cov}(PC_{t+1})]} \\ &= \sum_{i=1}^K \left(\frac{R_i^2}{1 - R_i^2} \right) \frac{\lambda_i}{\lambda}, \end{aligned} \quad (8)$$

where $\lambda = \sum \frac{\lambda_i}{1 - R_i^2} \approx \sum \lambda_i$ is the total unconditional variance of returns (Internet Appendix Section I shows the derivations). This quantity measures the total amount of predictability for the cross-section of returns, and has the useful feature to be invariant by rotation of the asset space. The second line shows that the total R^2 comes from the predictability of each of the PCs, weighted by their importance in explaining the factors. In the case of a single asset, the formula reduces to the standard predictive R^2 .

What happens when the set of portfolios exhibits a factor structure, that is some λ_i are large while others are smaller? Combining the total R^2 and maximum squared Sharpe ratio relations, one can see that small principal components cannot meaningfully contribute to predictability, or they would yield too high a Sharpe ratio. Intuitively, while it is entirely possible that each of the many proposed factors are predictable, it is unlikely they all

⁸ Gallant, Hansen, and Tauchen (1990) use this estimated unconditional variance to empirically test asset pricing models in the presence of conditioning information.

capture independent sources of risk. Otherwise, investors would be able to diversify across them and obtain implausibly large Sharpe ratios. This is the dynamic counterpart to the static reasoning of Kozak et al. (2018), who use a similar argument to conclude that small principal components cannot contribute meaningfully to the cross-sectional dispersion in *average* returns. Hence, the large few PC portfolios must capture both cross-sectional and time-series variation in expected factor returns. We define Z_{t+1} as the vector of the largest principal component portfolios of F_{t+1} . The exact number of PCs to include is an empirical question and depends on the strength of the factor structure. The following proposition summarizes the implications of this result for the SDF and the optimal factor timing portfolio.

Proposition 1. Under Assumption 1 and Assumption 2, the SDF can be approximated by a combination of the dominant factors:

$$m_{t+1} \approx 1 - \mathbb{E}_t[Z_{t+1}]' \Sigma_{Z,t}^{-1} (Z_{t+1} - \mathbb{E}_t[Z_{t+1}]). \quad (9)$$

Equivalently, the maximum Sharpe ratio factor timing portfolio can be approximated by

$$R_{t+1}^{\text{opt}} \approx \mathbb{E}_t[Z_{t+1}]' \Sigma_{Z,t}^{-1} Z_{t+1}. \quad (10)$$

Our two assumptions are complementary in the following sense. Assumption 1 delivers the conclusion that factor timing is sufficient; one need not time individual stocks. Even without this assumption, Assumption 2 provides a useful way to time a given set of factors. However, the two together allow us to measure properties of the SDF, shedding light on a fundamental economic quantity.

As we will show in the context of our empirical application, no more than a few principal components explain a sizable fraction of the variation in the factor returns. In addition, because factors are often chosen to offer a stable correlation structure, the extraction of dominant components is readily implementable using the standard unconditional method.⁹ We are left with estimating the conditional means and variances of these large principal components. In this paper, we concentrate on estimating the mean forecasts $\mathbb{E}_t[Z_{t+1}]$, that is produce return forecasts for a low-dimensional set of portfolio returns. The estimation of volatility is typically more straightforward, and we come back to it in Section 4.4.¹⁰ Because we only consider a few components, we use standard forecasting methods for individual returns.

⁹ Stock and Watson (2002) provide conditions under which unconditional principal components analysis identifies important components even in the presence of time-varying parameters.

¹⁰ Moreira and Muir (2017) and Engle and Kelly (2012) are examples of work that point to methods for timing volatility and the benefits it provides.

To summarize, our assumptions lead to the following approach to measure conditional expected returns and engage in factor timing:

1. Start from a set of pricing factors F_{t+1} .
2. Reduce this set of factors to a few dominant components, Z_{t+1} , using principal components analysis.
3. Produce separate individual forecasts of each of the Z_{t+1} , that is measures of $\mathbb{E}_t[Z_{t+1}]$.
4. To measure the conditional expected factor returns, apply these forecasts to factors using their loadings on the dominant components.
5. To engage in factor timing or estimate the SDF, use these forecasts to construct the portfolio given in Equation (10).

In the remainder of this paper, we implement this approach in the context of the cross-section of stock returns. We show how our method allows one to obtain robust measures of expected returns, which is useful for factor timing. And, by studying the corresponding SDF, we explain how it generates novel empirical facts to discipline economic models. In Internet Appendix Section III, we provide an alternative, more statistical motivation for our methodology.

2. Factor Return Predictability

2.1 Data

Step 1 of our approach is to start with a set of pricing factors. For equities, following the logic in Kozak et al. (2020), we focus on a broad set of fifty “anomaly” portfolios that effectively summarize the heterogeneity in expected returns. We present here the construction for our main estimates, and confirm the robustness of our conclusions around these choices in Table 4. We construct these portfolios as follows. We use the universe of CRSP and Compustat stocks and sort them into ten value-weighted portfolios for each of the fifty characteristics studied in Kozak et al. (2020) and listed in Internet Appendix Table A.3. Portfolios include all NYSE, AMEX, and NASDAQ firms; however, the breakpoints use only NYSE firms as in Fama and French (2016). Our sample consists of monthly returns from January 1974 to December 2017.¹¹

We construct the long-short anomalies as differences between each anomaly’s return on portfolio 10 minus the return on portfolio 1. For each anomaly strategy we also construct its corresponding measure of relative valuation based on book-to-market ratios of the underlying stocks. We define this measure, bm , as the difference in log book-to-market ratios of portfolio 10 and portfolio 1.¹²

¹¹ We make our data available at: <https://www.serhiykozak.com/data>

¹² The book-to-market ratio, bm , of a portfolio is defined as the sum of book equity relative to the total market capitalization of all firms in that portfolio. Equivalently, it is the market-capitalization weighted average of individual stocks’ bm ratios.

Table 1
Percentage of variance explained by anomaly PCs

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
% var. explained	25.8	12.4	10.3	6.7	4.8	4.0	3.6	2.8	2.2	2.1
Cumulative	25.8	38.3	48.5	55.2	60.0	64.0	67.6	70.4	72.6	74.7

This table reports the percentage of variance explained by each PC of the fifty anomaly strategies.

Most of these portfolio sorts exhibit a significant spread in average returns and capital asset pricing model (CAPM) alphas. This finding has been documented in the vast literature on the cross-section of returns and can be verified in Internet Appendix Table A.3. In our sample, most anomalies show a large, nearly monotonic pattern in average returns across decile portfolios, consistent with prior research. Rather than studying unconditional mean returns, our primary focus in this paper is on time variation in conditional expected returns. This is an area that has received considerably less attention in prior work.

Finally, we also market-adjust and rescale the data. Specifically, for each anomaly we compute its regression β with respect to aggregate market returns. We then market adjust our returns and predictors by subtracting $\beta \times r_{mkt}$ for returns and $\beta \times bm_{mkt}$ for the bm ratios. Next, we rescale the market-adjusted returns and bm ratios so that they have equal variance across anomalies. Importantly, the β s and variances used for these transformations are estimated using only the first half of the sample so that out-of-sample (OOS) statistics contain no look-ahead bias.

2.2 Dominant components of the factors

Step 2 of our approach is to reduce this set of factors to a few dominant components, its largest PCs. We are interested in the joint predictability of anomaly portfolio returns. We construct PCs from the fifty anomaly portfolios and study their predictability. Formally, consider the eigenvalue decomposition of anomaly excess returns, $\text{cov}(F_{t+1}) = Q \Lambda Q'$, where Q is the matrix of eigenvectors and Λ is the diagonal matrix of eigenvalues. The i -th PC portfolio is formed as $PC_{i,t+1} = q'_i F_{t+1}$ where q_i is the i -th column of Q . To ensure our later OOS results do not have any look-ahead bias, we estimate Q and Λ using only the first half of the data.

Table 1 shows that anomaly portfolio returns exhibit a moderately strong factor structure. For example, the first PC accounts for one fourth of the total variation. This is sizable but however, much weaker than what is typically found in other asset classes such as Treasury bonds or foreign exchange. How many components should we use? The relations of Equation (7) and (8) provide some guidance for this choice if we use some plausible priors. First, Campbell and Thompson (2007) show that the monthly R^2 when predicting the market is around 75 bp when using various price ratios, so we can use this as a reasonable magnitude for the predictability (total R^2) of the anomalies. Second, a relatively loose upper bound on the maximum Sharpe ratio $\mathbb{E}(S_t^2)$ is 1 at the annual

frequency—about twice that of the market—or 8.3% monthly. Under the view that all included PCs equally contribute to the total R^2 , the harmonic mean of their contribution to the total variance of returns must be higher than the ratio of these two numbers, $75 \text{ bp}/8.3\% = 9\%$.¹³ Using the eigenvalues in Table 1, this value yields that we should include at most five PCs. Based on this simple calculation, we focus on these for our main analysis and explore robustness to other choices. These five components jointly explain nearly two-thirds of the total variation in returns. Our portfolios are market neutral, so we also include the aggregate market portfolio as a potentially important pricing factor. In other words, we study $Z_{t+1} = (R_{mkt,t+1}, PC_{1,t+1} \cdots PC_{5,t+1})$.

2.3 Predicting the large PCs of anomaly returns

Step 3 of our approach is to produce individual forecasts of the dominant components of factor returns.

2.3.1 Predictors. We obtain these forecasts using standard predictive regressions on valuation ratios. Valuation ratios are the most commonly used forecasters for the market return, going back to Shiller (1981), Fama and French (1988), and Campbell and Shiller (1988). They also have been used at the individual stock level, by Lochstoer and Tetlock (2020) and Vuolteenaho (2002). Cohen, Polk, and Vuolteenaho (2003) show that value-minus-growth strategies are predictable by their own book-to-market ratios. Kelly and Pruitt (2013) use a large cross-section of book-to-market ratios to predict both the aggregate market return as well as portfolios sorted on various characteristics. This broad use comes from the fact that one should expect them to be informative about expected returns. For example, log-linearizing the clean surplus accounting relation of Ohlson (1995), Vuolteenaho (2002) shows that the log book-to-market ratio of a long-only strategy is a discounted sum of all future expected returns for this strategy minus future earning growth. By using valuation ratios as well, our conclusions are readily comparable to this seminal work. However, we are not arguing that other predictors, perhaps motivated by specific economic theories, could not find additional sources of predictability.

Following our broad goal of dimension reduction, we construct a single predictor for each portfolio: **we use its net book-to-market ratio**. For predicting $PC_{i,t+1}$, we construct its own log book-to-market ratio $bm_{i,t}$ by combining the anomaly log book-to-market ratios according to portfolio weights: $bm_{i,t} = q'_i bm_t^F$. We use the difference between this quantity for the long and short leg of our PCs, thereby capturing potentially useful information about future expected returns. Intuitively, $bm_{i,t}$ keeps track of the relative valuation of stocks of a certain *fixed* combination of “types” (e.g., value vs. growth, large vs. small), where these “types” are varying across stocks. When the valuation spread

¹³ Internet Appendix Section I provides a derivation of this formula.

Table 2
Predicting dominant equity components with BE/ME ratios

	MKT	PC1	PC2	PC3	PC4	PC5
Own <i>bm</i>	0.76 (1.24)	4.32 (4.31)	1.62 (1.81)	1.80 (2.01)	4.86 (3.74)	1.56 (0.78)
bias	0.68	0.36	0.16	0.18	0.10	0.08
<i>p</i> -value	.35	.00	.10	.07	.00	.48
<i>R</i> ²	0.29	3.96	0.74	0.56	3.59	0.50
OOS <i>R</i> ²	1.00	4.82	0.97	0.47	3.52	0.55
OOS <i>R</i> ² critical values						
90th	0.44	0.49	0.29	0.21	0.71	0.59
95th	0.68	0.97	0.48	0.37	1.19	0.96
99th	1.35	1.73	0.87	0.84	1.95	1.71

We report results from predictive regressions of excess market returns and five PCs of long-short anomaly returns. The market is forecasted using the log of the aggregate book-to-market ratio. The anomaly PCs are forecasted using a restricted linear combination of anomalies' log book-to-market ratios with weights given by the corresponding eigenvector of pooled long-short strategy returns. The first row shows the coefficient estimate. The second row shows asymptotic *t*-statistics estimated using the method of Newey and West (1987). The third and fourth rows show the bias and *p*-value from a parametric bootstrap. The fifth and sixth rows shows the in-sample and out-of-sample monthly *R*². The last three rows give critical values of the OOS *R*² based on the placebo test in Kelly and Pruitt (2013).

is large, it is natural to expect that stocks of the corresponding combination of “types” will experience low future returns. These portfolio-level book-to-market ratios are likely to be stable (even though portfolio composition is changing) because expected returns depend on this combination of “types”; empirically we find they are stable.

In addition, note that this choice of predictors dramatically reduces the dimensionality of the set of predictive variables. In Section 2.6 we explore alternative methods for analyzing high-dimensional data.

2.3.2 Predictability results. We analyze the predictability of anomaly PC portfolios and the aggregate market using a monthly holding period, as in Campbell and Thompson (2007). Table 2 shows the results of these six predictive regressions. The first two rows report the predictive coefficient estimate and Newey and West (1987) *t*-statistic.¹⁴ The third and fourth rows show the bias in coefficient estimate and *p*-value obtained from a parametric bootstrap. **Precisely, we first estimate a restricted VAR(1) for a PC's return and *bm* ratio under the null of no predictability.** We then simulate 10,000 histories from the VAR system with errors drawn with replacement from the empirical distribution. From these simulations, we obtain the distribution of coefficients and *t*-statistics. We construct *p*-values by using the simulated distribution of the *t*-statistics. Both the asymptotic standard errors and these *p*-values are useful for inference: the Newey and West (1987) standard errors is consistent under mild assumptions on the data-generating process, while the *p*-value corrects

¹⁴ We use a 2-year window for the Bartlett kernel.

for potential finite-sample biases and nonnormality. The fifth row gives the full sample predictive R^2 and the sixth row reports the OOS R^2 .¹⁵ To compute this statistic, we divide the sample into two equal halves. We estimate predictive coefficients in the first half and apply these to bm ratios in the second half to form OOS forecasts. All data construction choices use only the first half data, so our OOS results are not subject to look-ahead biases.

Consistent with previous studies, the estimate is not statistically significant for the market. While the OOS R^2 of 1% is encouraging, the bootstrap exercise reveals that there is substantial Stambaugh bias, almost as large as the estimated coefficient. In contrast, PCs 1 and 4 are unambiguously predictable by their own bm ratio. The estimated coefficients are large and significant, with t -statistics around 4 and bootstrapped p -values close to 0. Both the in-sample and OOS R^2 s are large: around 4% for PC1 and around 3.5% for PC4. The estimated relation for PC2 and PC3 exhibit weaker strength, but still appears statistically significant. t -statistics are 1.81 and 2.01, whereas bootstrapped p -values are 11% and 7%. The OOS R^2 s take values around 1% and 0.5%. Finally, PC5 does not appear predictable, with an insignificant coefficient estimate. For these PC portfolios, coefficients are slightly biased upward. In contrast to the market estimates, the biases for the PC portfolios are small relative to the estimated coefficients. In Internet Appendix Section II we show that this substantially lower bias for the PC portfolios obtains for two reasons. First, their bm ratios are less persistent than that of the aggregate market. Second, the correlation of innovations to bm ratios and returns is lower for the PCs than for the market.

To further demonstrate that the predictability we uncover does not arise due to mechanical biases, we run placebo tests following Kelly and Pruitt (2013a). This allows us to assess whether the OOS R^2 s we obtain are statistically significant. Specifically, we generate six AR(1) processes that are specified to have the same mean, autocorrelation, and covariance as the bm ratios we use as predictors.¹⁶ Because these are simulated values, they are independent of true return data. We then construct OOS forecasts for actual returns using the simulated data and record the OOS R^2 values. We repeat this procedure 1,000 times to obtain simulated distributions for OOS R^2 statistics. From these distributions, we compute 90%, 95%, and 99% critical values, which are reported in the last three rows of Table 2. Comparing the estimated OOS R^2 with the critical values reinforces the conclusions we drew above. The estimated OOS R^2 for the PCs are unlikely to be due to mechanical finite-sample biases: they are all significant at the 5% level except for PC5.

In Internet Appendix Figure A.1, we report the time series of realized returns along with full sample and out-of-sample forecasts of returns on the aggregate market and the first five PC portfolios of anomalies. In particular, for PC1 and

¹⁵ We define the OOS R^2 as $1 - \frac{\text{var}(r - \hat{r})}{\text{var}(r)}$, where $\hat{r}$ is the forecast formed using parameters estimated in sample.

¹⁶ We match these moments using the first half of the data to avoid any potential look-ahead bias.

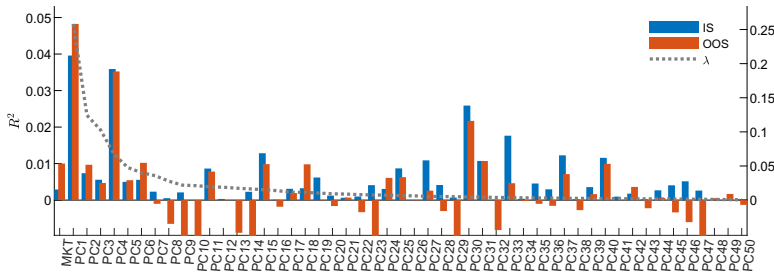


Figure 1
Predictability of equity PCs with own *bm* ratios

The plot shows the in-sample (IS) and out-of-sample monthly R^2 of predicting each PC of anomaly portfolio returns with its own *bm* ratio. The dotted gray line (using the right axis) represents the fraction of total variance explained by each PC.

PC4, the out-of-sample forecasts are remarkably close to the full sample values, further confirming that our coefficient estimates are precisely measured.

2.3.3 Importance of restrictions. Our emphasis on the largest principal components is predicated on the idea that they should capture most of the variation in expected returns. In Figure 1, we report predictability for *all* the principal components of our factors. Only large PCs are strongly predictable in- and out-of-sample by their own *bm* ratios. In this sense, our focus on predicting only the first few PCs is not only motivated by theory, but finds strong support in the data. In addition, by focusing on predicting only these dominant components we uncover robust predictability in the data and largely ignore spurious predictability that stems from small PCs. This latter result echoes the more statistical concerns we put forward as well.

We now turn to forecasting individual anomalies to consider how the restrictions we impose help in this exercise.

2.4 Predicting individual factors

Step 4 of our approach is to infer expected return forecasts for the individual factors from the forecasts of our dominant components. Factors are known linear combinations of the PC portfolios, so we can use the estimates in Table 2 to generate forecasts for each anomaly. Notably, each anomaly return is implicitly predicted by the whole cross-section of *bm* ratios. Table 3 shows the in- and out-of-sample R^2 for each anomaly return using our method. Many of the anomalies are highly predictable; roughly half have OOS R^2 greater than 1% and only two have OOS R^2 below -1% . The total R^2 is 1.03% in-sample and 0.93% OOS.¹⁷ Our approach allows us to uncover these patterns in a robust way.

¹⁷ We measure the OOS total R^2 as $1 - \frac{\text{tr}[\text{cov}(\varepsilon_{t+1})]}{\text{tr}[\text{cov}(F_{i,t+1})]}$, where ε_{t+1} are forecast errors. This quantity can be negative, which typically obtains if forecasts and realizations are negatively correlated out of sample.

Table 3
Predicting individual anomaly returns: R^2 (%)

	IS	OOS		IS	OOS
1. Size	3.8	4.5	28. Short interest	-0.5	-0.4
2. Value (A)	1.8	1.9	29. Momentum (12m)	1.3	1.4
3. Gross profitability	-2.2	-4.7	30. Industry momentum	0.1	-0.2
4. Value-profitability	3.7	3.8	31. Momentum-reversals	2.8	3.3
5. F-score	0.4	-0.2	32. Long run reversals	5.7	5.5
6. Debt issuance	0.8	0.5	33. Value (M)	3.6	3.0
7. Share repurchases	0.9	0.7	34. Net issuance (M)	0.8	1.3
8. Net issuance (A)	2.4	3.8	35. Earnings surprises	-0.9	-0.7
9. Accruals	-0.2	-0.1	36. Return on book equity (Q)	0.2	0.0
10. Asset growth	1.8	2.6	37. Return on market equity	0.7	0.1
11. Asset turnover	0.6	0.8	38. Return on assets (Q)	0.3	0.5
12. Gross margins	0.6	-1.0	39. Short-term reversals	0.3	0.5
13. Earnings/price	0.7	-0.0	40. Idiosyncratic volatility	1.5	0.6
14. Cash flows/price	0.7	0.4	41. Beta arbitrage	-0.6	-0.6
15. Net operating assets	0.6	-0.2	42. Seasonality	-0.4	0.1
16. Investment/assets	1.8	1.3	43. Industry rel. reversals	1.2	1.0
17. Investment/capital	-0.1	-0.5	44. Industry rel. rev. (L.V.)	1.9	1.6
18. Investment growth	1.8	1.9	45. Ind. mom-reversals	0.6	-0.2
19. Sales growth	1.2	2.2	46. Composite issuance	-0.3	0.1
20. Leverage	0.6	0.7	47. Price	4.3	3.5
21. Return on assets (A)	0.9	1.2	48. Share volume	-0.4	-0.6
22. Return on book equity (A)	1.2	-0.1	49. Duration	2.1	2.9
23. Sales/price	2.0	1.2	50. Firm age	0.3	0.5
24. Growth in LTNOA	0.5	0.7			
25. Momentum (6m)	1.7	1.7	Mean	1.1	1.0
26. Value-momentum	-0.0	1.2	Median	0.8	0.7
27. Value-momentum-prof.	1.7	2.5	SD	1.4	1.7

Monthly predictive R^2 of individual anomalies returns using implied fitted values based on PC forecasts. Column 1 (IS) provides estimates in the full sample. Column 2 (OOS) shows out-of-sample R^2 .

The substantial anomaly predictability we document in Table 3 also contributes to the recent debate on whether these strategies represent actual investment opportunities or are statistical artifacts, which are largely data mined. For example, Hou, Xue, and Zhang (2018) claim that most anomalies are not robust to small methodological or sample variations and conclude there is “widespread p -hacking in the anomalies literature.” Using a different methodology, Harvey and Liu (2016) argue that the three-factor model of Fama and French (1993) provides a good description of expected returns, and, hence, most CAPM anomalies are spurious. Interestingly, we find that some anomalies, such as size and sales growth, which have low unconditional Sharpe ratios are nonetheless highly predictable. This indicates that these strategies at least sometimes represent “important” deviations from the CAPM. This echoes the importance of conditioning information, as emphasized by Nagel and Singleton (2011) and others. More generally these results highlight that a lack of risk premium on average does not necessarily imply a lack of risk premium always: expected returns can be sometimes positive and sometimes negative for the same strategy.

Table 4
Various data choices

Portfolio sort	Holding period (months)	# of PCs	Monthly PCs	Market adjusted returns	Scaled returns & <i>bm</i>	OOS Total R^2	# PCs signif. R^2_{OOS}
Deciles	1	5	X	X	X	0.93	4
Quintiles	1	5	X	X	X	1.01	4
Terciles	1	5	X	X	X	0.81	3
Deciles	1	1	X	X	X	0.57	1
Deciles	1	2	X	X	X	0.72	2
Deciles	1	3	X	X	X	0.78	3
Deciles	1	4	X	X	X	0.90	4
Deciles	1	6	X	X	X	0.97	5
Deciles	1	7	X	X	X	0.96	5
Deciles	1	5	X	X		1.18	3
Deciles	1	5	X		X	1.27	4
Deciles	1	5	X			1.24	2
Deciles	3	5	X	X	X	2.69	4
Deciles	3	5		X	X	0.99	2
Deciles	6	5	X	X	X	5.42	3
Deciles	6	5		X	X	3.49	3
Deciles	12	5	X	X	X	10.05	3
Deciles	12	5		X	X	4.54	2

The table reports summary statistics of predictive regressions in Table 2 for various data construction choices. Specifically, we report the OOS total R^2 and the number of PC portfolios for which the OOS R^2 is statistically significant using the placebo test of Kelly and Pruitt (2013a). The first column reports the number of portfolios used for the underlying characteristic sorts. The second column reports the holding period in months. For holding periods longer than 1 month, the third column reports whether principal components are estimated using monthly or holding period returns. The fourth column reports whether the anomaly returns are orthogonalized relative to the aggregate market. The fifth column reports whether the anomaly returns and book-to-market values are normalized to have equal variance.

2.5 Robustness

Our main estimation includes many choices such as how to construct the raw anomaly returns, the length of the holding period, how many PCs to include, whether to market-adjust, and whether to rescale the data. In Table 4, we explore the robustness of our results to changes in these specifications. For each specification, we report the OOS total R^2 as well as the number of PCs with statistically significant OOS R^2 at the 5% level. The first row repeats the results for our main specification. The next two rows show that our results are robust to how the anomaly portfolios are constructed. Instead of first sorting stocks into deciles for each characteristics, we use quintiles and terciles and obtain similar results. In the next block, we consider varying the number of included PCs from one to seven and again obtain similar OOS findings. Adding more components does not meaningfully enhance performance. Reducing the number of PCs below four leads to some reduction in performance. Still, with only one PC we obtain more than half the OOS R^2 of our baseline model. Next, we consider not market adjusting or rescaling returns and *bm* ratios. In fact, the OOS total R^2 improves without these transformations. Finally, we consider different holding periods. With quarterly, half-year, and annual holding periods, the OOS total R^2 increases almost proportionately with horizon. Importantly, this only obtains when estimating the PC eigenvectors using monthly returns.

That is, even with a 12-month holding period, we can construct the PCs using the covariance matrix of monthly holding period returns. If, instead, we construct PCs using only returns at the same frequency, meaningful information about the covariance structure of returns is lost. We use nonoverlapping returns, so increasing the holding period proportionately reduces the sample size. For an annual holding period, the eigenvectors are estimated using only twenty-two observations so the sample covariance matrix is not even of full rank. It is evidently beneficial to use higher frequency data to estimate the covariance matrix and resultant eigenvectors. In Internet Appendix Section II we show that our results are robust to using expanding- and rolling-window OOS methods, as well as estimating the regressions in the second half of the sample and measuring OOS performance in the first half.

2.6 Comparison to alternative approaches

Forecasting only large PC portfolios by their own *bm* ratio generates robust OOS total R^2 , but one could forecast anomaly returns in many other ways. Some papers advocate methods that aim to deal with the high dimensionality of either the forecast targets (returns) or the predictors (*bm* ratios). Others use different predictive variables altogether. In Table 5 we report the OOS total R^2 as well as mean, median, and standard deviation of individual anomaly R^2 s for a wide variety of these approaches. All statistics are out-of-sample. The first row shows results from a completely unregularized estimation in which each anomaly return is forecast using all fifty anomaly *bm* ratios. As expected, the OOS total R^2 of -134% is terrible. This highlights the need for dimensionality reduction. The second row reports results from our method.

Instead of predicting each PC return with only its own *bm*, we could expand the information set and allow each PC return to be forecast by any or all of the five PCs' *bm* ratios. Because an ordinary least squares (OLS) regression with even five time-series predictors is likely substantially overfit, we consider various regularization schemes. We first consider ridge regression and Lasso estimation with penalty parameters chosen to allow for exactly one degree of freedom, as in our main estimation. Row 3 shows that ridge regression, even with five predictors, does not deliver robust predictability. Row 4 shows that Lasso does somewhat better, but still substantially worse than using only each portfolio's own *bm*. Belloni and Chernozhukov (2013) show theoretically that the "OLS post-Lasso estimator performs at least as well as Lasso in terms of the rate of convergence, and has the advantage of a smaller bias." Row 5 confirms this result empirically. Using Lasso to select one predictor, then estimating the coefficient by OLS produces a 0.76% total R^2 , nearly as high as our method. Instead of using the *bm* ratios of the PC portfolios, we could have used PCA directly on the anomaly *bm* ratios to reduce the dimensionality of the predictors. Row 6 shows that OLS post-Lasso on the five principal components of anomaly *bm* ratios does reasonably well, but worse than using the *bm* ratios of the PCs themselves. Price ratios are much more persistent than returns, so the sample

Table 5
Out-of-sample R^2 of various forecasting methods

Method	OOS total R^2	Mean	Median	SD
1. 50 Anom, BM of Anom, OLS	-133.73	-161.91	-123.12	129.75
2. 5 PCs, Own BM	0.93	1.00	0.69	1.69
3. 5 PCs, BM of PCs, Ridge 1DoF	0.01	0.02	0.02	0.09
4. 5 PCs, BM of PCs, Lasso 1DoF	0.26	0.27	0.19	0.56
5. 5 PCs, BM of PCs, Lasso-OLS 1DoF	0.76	0.83	0.61	1.75
6. 5 PCs, PCs of BM, Lasso-OLS 1DoF	0.52	0.59	0.35	1.17
7. 5 PCs, BM of PCs, 3PRF	0.32	0.36	0.19	0.96
8. 50 Anom, BM of Anom, Lasso-OLS 1DoF	-2.79	-3.27	-1.04	5.10
9. 50 Anom, BM of PCs, Lasso-OLS 1DoF	0.03	-0.06	-0.18	2.33
10. 50 Anom, Own BM	0.50	0.49	0.11	1.42
11. 50 Anom, Own BM, Pooled	0.48	0.51	0.42	1.13
12. 50 Anom, BM of Anom, 3PRF	0.16	0.17	0.12	0.81
13. 50 Anom, Own Characteristic	-2.94	-3.21	0.03	20.67
14. 50 Anom, Sentiment	0.17	0.06	0.01	1.19
15. 5 PCs, Sentiment	0.19	0.06	0.01	1.19
16. 50 Anom, Factor Momentum	-0.49	-0.48	-0.32	1.12
17. 5 PCs, Factor Momentum	-0.08	-0.05	-0.23	1.19

The table reports the monthly OOS total R^2 as well as mean, median, and standard deviation of OOS R^2 for individual anomaly portfolios for various forecasting methods. The first column gives the set of assets that are directly forecast, the predictive variables used, and the forecasting method. When omitted, the method is ordinary least squares.

covariance matrix and resultant eigenvectors are measured with substantially more error, which may contribute to the worse performance. Row 7 uses the five PCs' *bm* ratios, but uses the three-pass regression filter (3PRF) from Kelly and Pruitt (2013) instead of ridge or Lasso. As in that paper, the filter is run separately for each dependent variable (PC return). For this set of portfolios, 3PRF does produce moderate OOS predictability, but less than OLS post-Lasso.

Rows 8 to 12 consider various methods for reducing the dimensionality of the predictors but without any left-hand side (LHS) dimensionality reduction. These methods predict each of the fifty anomaly returns directly. Row 8 shows that OLS post-Lasso using fifty *bm* fails completely OOS. Instead of using the fifty anomaly *bm* ratios, we could use the five PCs' *bm* ratios as in row 5. This approach does substantially better (row 9), improving to a near-zero OOS total R^2 . However, row 5 shows that with the same information set and same estimation technique, restricting to PCs of returns produces a large OOS total R^2 , highlighting the importance of reducing the dimension of the LHS.

In row 10, we predict each anomaly with its own *bm* ratio, as our method does for PC portfolios. Perhaps not surprisingly, this produces a substantial total R^2 of 0.5%. Price ratios seem to be robust predictors of returns. Still, the total R^2 is only slightly greater than half of what we find when directly predicting large PC portfolios. This highlights that while an asset's price ratio may be informative about its expected return, there is valuable information in the whole cross-section of valuation ratios, a point first made by Kelly and Pruitt (2013a). A number of previous papers use a similar methodology. Asness et al. (2000) and Cohen, Polk, and Vuolteenaho (2003) find that the

value anomaly itself is forecastable by its own *bm* ratio.¹⁸ Arnott, Beck, and Kalesnik (2016b) use various valuation measures (price ratios) to forecast eight anomaly returns; they forecast each anomaly return with its own valuation measures and find statistically significant results. Asness et al. (2017) use each anomaly's *bm* ratio to construct timing strategies for the value, momentum, and low β anomalies. Using their methodology, however, they conclude that the strength of predictability is lacking. Implicitly, these methods allow for as many independent sources of time-varying expected returns as there are assets or factors to predict.¹⁹ This framework is optimal only if one thinks the anomalies and their expected returns are independent, at odds with our Assumption 2.

In row 11, we show results from repeating the exercise of forecasting each anomaly with its own *bm* ratio, but now estimate the predictive coefficients from a panel regression that imposes that the coefficient on *bm* is the same for all anomalies. Formally, we estimate $\mathbb{E}_t(F_{i,t+1}) = a_{0,i} + a_1 bm_{i,t}$, allowing each anomaly factor to have a different intercept, thereby allowing different unconditional means, but imposing uniform predictability of returns using *bm*. The OOS performance is not meaningfully different from unrestricted estimation. In related work, Campbell, Polk, and Vuolteenaho (2009) and Lochstoer and Tetlock (2020) use a bottom-up approach of aggregating firm-level estimates to portfolios in order to decompose variation in returns into discount rate and cash-flow news. They estimate a panel VAR in which they forecast each stock's return using its own *bm* ratio, additional stock characteristics, and some aggregate variables.²⁰ Unlike the previous studies, their study imposes the restriction that the coefficients in the predictive regression be the same for all stocks. Although imposing this equality is a form of statistical regularization, it still allows for as many independent sources of time-varying expected returns as there are stocks. Therefore, these restrictions do not discipline the Sharpe ratios implied by the predictability estimates.

In row 12, we directly apply the 3PRF to the anomaly returns, using their fifty *bms* as predictors. This is unlike row 7, where we first reduce by the LHS dimension before estimating with 3PRF. As with PCs, 3PRF generates a moderate OOS total R^2 , but substantially less than just using an asset's own *bm* ratio. These lower R^2 s for the 3PRF might seem at odds with the numbers reported in Kelly and Pruitt (2013a). One reason for this difference is that we work with market-neutral long-short portfolios whereas that paper focuses on long-only characteristic-sorted portfolios, all of which have a market β of

¹⁸ Asness, Friedman, Krail, and Liew (2000) use the ratio of the book-to-market of value stocks to that of growth stocks. Cohen, Polk, and Vuolteenaho (2003) uses the log difference, as we do.

¹⁹ Formally, the covariance matrix of expected returns is allowed to have full rank.

²⁰ Campbell, Polk, and Vuolteenaho (2009) do not include any aggregate variables and further cross-sectionally demean stock returns each period. Most stocks have a market β close to unity, so their VAR approximately forecasts market-neutral stock returns using each stock's own *bm* ratio and other characteristics.

around unity. The PLS implementation in Kelly and Pruitt (2013) assumes that each target variable (asset return) loads on only one of the (possibly) few latent factors that drive the cross-section of expected returns, cash flow growth rates, and valuation ratios. 3PRF is successful at predicting the aggregate market, so it should perform nearly as well at predicting these other portfolios, because they all approximately equal the market return plus some smaller other source of variation. Their β is close to 1, and they are highly correlated to the market. This conclusion holds even if these other variations are unpredictable, as shown in Internet Appendix Section III. This explanation can rationalize why their cross-sectional portfolios are well predicted by 3PRF, but not our long-short factor portfolios. In fact, their findings for size-sorted portfolios are consistent with this view. The OOS R^2 s increase monotonically with firm size, increasing from -18% for the smallest quintile, which is less correlated with the value-weighted market, to $+12\%$ for the largest quintile, which is strongly correlated with the market.

Light, Maslov, and Rytchkov (2017) develop a related approach to construct forecasts for each stock each month using a large set of characteristics. However, their method does not provide a forecast $\mu_{i,t}$ but rather a scaled forecast $F_t \mu_{i,t}$, where the scaling factor F_t is unobserved and has arbitrary time-varying scale. Hence, their method, while confirming the existence of timing benefits, cannot be directly compared to other forecasting approaches.

In rows 13–17, we consider predictors besides *bm* ratios. In row 13, we present a natural alternative: predicting each anomaly with its own characteristic spread. For example, when sorting firms into deciles based on log market capitalization, the characteristic spread at time t is $\log(ME_{1,t}) - \log(ME_{10,t})$ where $ME_{i,t}$ is the weighted average market capitalization of firms in decile i at time t . As above with *bm* ratios, each anomaly is forecast using its own anomaly specific variable. Somewhat surprisingly, this approach does quite poorly. The performance could possibly be improved through judicious transformation of the sorting variable, though without theoretical guidance on the functional form such an exercise is prone to data mining concerns. In a related paper, Greenwood and Hanson (2012) forecast characteristic-based anomalies using their “issuer-purchaser” spread, or the difference in characteristic (say *ME* or *bm*) for net equity issuers versus repurchasers. However, as above, a concern for approaches using an anomaly-specific forecasting variable implicitly is that it allows for as many independent sources of time-varying expected returns as there are assets or factors to predict.

There is another literature quite unlike the above alternatives, focusing on one or a few return predictors for all anomalies together. Stambaugh, Yu, and Yuan (2012) forecast twelve anomaly strategy returns using the aggregate sentiment index of Baker and Wurgler (2006) and find statistically significant predictability for most of the anomalies they consider. Further, the predictive coefficients are of similar magnitude across anomalies. This is reminiscent of the tradition in bond return forecasting that seeks a single variable that is

a significant predictor of excess bond returns of all maturities (Cieslak and Povala 2015; Cochrane and Piazzesi 2005, 2008). Using a single predictor for all assets implicitly or explicitly assumes there is only a single source of time-varying expected returns.²¹ Stambaugh, Yu, and Yuan (2012) do not report R^2 statistics, but a back-of-the-envelope calculation gives values ranging from 0.5%–4% across the anomalies. Rows 14 and 15 show that sentiment does produce moderate OOS predictability, but much less than in-sample. This is true even if we restrict to predicting the five PC portfolios. This suggests that, although sentiment is an important variable in estimating variation in expected returns across anomalies, it captures only a small fraction of this total variation. Akbas et al. (2015) start similarly, forecasting eleven anomaly returns individually using aggregate mutual fund and hedge fund flows. Based on the pattern of coefficients, they divide the anomalies into two groups: “real investment factors” and “others.” They then form an aggregate portfolio return for each group and forecast these two returns and find substantial R^2 for the “other group” and nearly zero for the investment group. Finally, Ehsani and Linnainmaa (2019) show that for fifteen anomalies, the anomaly’s own prior performance significantly positively predicts its return in month t . Rows 16 and 17 show that for our broader set of anomalies, “factor momentum” does not predict returns OOS, with R^2 s of -0.49% and -0.08% .²²

3. Optimal Factor Timing Portfolio

We turn to step 5 of our approach: using our forecasts to form an optimal factor timing portfolio. We find large benefits to factor timing, benefits that appear feasible to collect in practice. This portfolio is also of interest economically, because it informs the properties of the SDF, which we discuss in Section 4.

3.1 Performance

The strong evidence of predictability we document yields substantial investment benefits. By scaling up positions when expected returns are larger, an investor can increase the performance of her portfolio. For example, consider the case of one asset with time-varying expected excess return μ_t and constant variance σ^2 . In this situation, the optimal portfolio of a mean-variance investor invests proportionally to μ_t/σ^2 at each point in time. This position generates a certainty equivalent proportional to $(\mathbb{E}[\mu_t]^2 + \text{var}[\mu_t])/\sigma^2$.²³ The second term, $\text{var}[\mu_t]/\sigma^2$, is the gain from taking advantage of variation in expected returns.

²¹ Formally, the covariance matrix of expected returns has at most rank one.

²² We follow Ehsani and Linnainmaa (2019) and forecast each portfolio with an indicator variable that equals one if the portfolio’s average monthly return over the past year is positive and zero otherwise.

²³ For an investor with risk aversion γ , the proportionality constant is $1/\gamma$ for the portfolio weight and $1/(2\gamma)$ for the certainty equivalent. Campbell and Thompson (2007) and Moreira and Muir (2017) discuss similar utility calculations.

Table 6
Performance of various portfolio strategies

	Factor investing	Market timing	Factor timing	Anomaly timing	Pure anom. timing
IS Sharpe ratio	1.27	1.23	1.19	1.19	0.71
OOS Sharpe ratio	0.76	0.63	0.87	0.96	0.77
IS inf. ratio	—	−0.17	0.36	0.37	0.35
OOS inf. ratio	—	−0.64	0.42	0.60	0.59
Expected utility	1.66	1.69	2.96	2.92	1.26

The table reports the unconditional Sharpe ratio, information ratio, and average mean-variance utility of five strategies: (1) a static factor investing strategy, based on unconditional estimates of $\mathbb{E}[Z_t]$; (2) a market timing strategy, which uses forecasts of the market return based on Table 2 but sets expected returns on the PC equal to unconditional values; (3) a full factor timing strategy including predictability of the PCs and the market; (4) an anomaly timing strategy, which uses forecasts of the PCs based on Table 2 but sets expected returns on the market to unconditional values; and (5) a pure anomaly timing strategy, which sets the weight on the market to zero and invests in anomalies proportional to the deviation of their forecast to its unconditional average, $\mathbb{E}_t[Z_{t+1}] - \mathbb{E}[Z_t]$. All strategies assume a homoscedastic conditional covariance matrix, estimated as the covariance of forecast residuals. Information ratios are calculated relative to the static strategy. Out-of-sample (OOS) values are based on split-sample analysis with all parameters estimated using the first half of the data.

If the same investor invests in a static portfolio with constant weight $\mathbb{E}[\mu_t]/\sigma^2$, she would only obtain the first term. With multiple assets, the optimal portfolio is given by Equation (10), which yields gains given by Equation (7). Here again, one can see that both the mean and variation in average returns increase the Sharpe ratio. To quantify the gains to factor timing in our setting, we implement this portfolio. Specifically, we use our method to predict component means, $\mathbb{E}_t[Z_{t+1}]$. We then use these forecasts to construct forecast errors and compute an estimate of the conditional covariance matrix of the market and PC returns, $\Sigma_{Z,t}$, which we for now assume is homoscedastic in order to estimate the role of forecasting means; we revisit this assumption in Section 4.4. We combine these estimates into portfolio weights $\omega_t = \Sigma_{Z,t}^{-1} \mathbb{E}_t[Z_{t+1}] = \Sigma_Z^{-1} \mathbb{E}_t[Z_{t+1}]$. Importantly, remember that the components Z_{t+1} are fixed linear combinations of the factors F_{t+1} . So, although we express the factor timing portfolio in terms of the components, it is implicitly trading the underlying factors.

Table 6 reports measures of the performance for versions of this portfolio under various assumptions. We consider five variations of the optimal timing portfolio. “Factor timing” (F.T.) is the portfolio described above. “Factor investing” (F.I.) sets all return forecasts to their unconditional mean, while “market timing” (M.T.) does the same except for the market return. “Anomaly timing” (A.T.) does the opposite: the market is forecast by its unconditional mean, while anomalies receive dynamic forecasts. Finally, the “pure anomaly timing” (P.A.T.) portfolio sets the weight on the market to zero and invests in anomalies proportional to the deviation of their forecast to its unconditional average. In other words, this portfolio has zero average loading on all factors, and lets us zoom in on the new information of this paper: variation in anomaly expected returns. The following equations summarize these strategies:

$$\omega_{\text{F.T.},t} = \Sigma_Z^{-1} \left[\mathbb{E}_t(R_{mkt,t+1}), \mathbb{E}_t(PC_{1,t+1}), \dots, \mathbb{E}_t(PC_{5,t+1}) \right]', \quad (11)$$

$$\omega_{\text{F.I.}} = \Sigma_Z^{-1} [\mathbb{E}(R_{mkt,t+1}), \mathbb{E}(PC_{1,t+1}), \dots, \mathbb{E}(PC_{5,t+1})]', \quad (12)$$

$$\omega_{\text{M.T.,}t} = \Sigma_Z^{-1} [\mathbb{E}_t(R_{mkt,t+1}), \mathbb{E}_t(PC_{1,t+1}), \dots, \mathbb{E}_t(PC_{5,t+1})]', \quad (13)$$

$$\omega_{\text{A.T.,}t} = \Sigma_Z^{-1} [\mathbb{E}(R_{mkt,t+1}), \mathbb{E}_t(PC_{1,t+1}), \dots, \mathbb{E}_t(PC_{5,t+1})]', \quad (14)$$

$$\omega_{\text{P.A.T.,}t} = \Sigma_Z^{-1} [0, [\mathbb{E}_t - \mathbb{E}](PC_{1,t+1}), \dots, [\mathbb{E}_t - \mathbb{E}](PC_{5,t+1})]'. \quad (15)$$

The first performance metric we consider is the *unconditional* Sharpe ratio: the ratio of the sample mean and the standard deviation of returns. The factor investing, market timing, anomaly timing, and factor timing portfolio all produce meaningful performance, with Sharpe ratios around 1.2 in sample and between 0.63 and 0.87 out of sample. One might be tempted to conclude from these numbers that factor timing does not improve performance relative to static factor investing. However, it is important to remember that the factor timing portfolio is not designed to maximize the unconditional Sharpe ratio. In a world with predictability, this is not an accurate measure of performance improvement. Ferson and Siegel (2001) show that maximizing the unconditional Sharpe ratio requires portfolio weights that are highly nonlinear and *nonmonotone* in conditional expected returns. Still, the sizable Sharpe ratio of the pure anomaly timing portfolio is a first piece of evidence that factor timing is valuable. This portfolio does not engage in any static bets, but obtains Sharpe ratios of 0.71 and 0.77 in and out of sample.

A second way to evaluate the value of factor timing is to assess whether the timing portfolios expand the unconditional investment opportunity set captured by the static factor investing portfolio. To do so, we compute the information ratio for the various timing strategies. This number corresponds to the Sharpe ratio that these strategies produce once orthogonalized to the static factor investing portfolio. An advantage of this statistic is that it can be measured without relying on assumptions about return dynamics. However, it only constitutes a *lower bound* on the benefits to factor timing, because it takes the perspective of an uninformed investor. Judged by the information ratio, factor timing, anomaly timing, and pure anomaly timing substantially extend the investment opportunity set. Their information ratios are almost equal at 0.36 in sample, and actually increase to 0.42, 0.60, and 0.59 out of sample. These stable information ratios, in contrast to the more strongly decaying Sharpe ratio of the factor investing portfolio suggests that our conclusions about predictability are actually more robust than the measurement of unconditional return premiums.

A third take on the value of factor timing is to ask, given our completely estimated model, what utility would a mean-variance investor expect to obtain on average.²⁴ Absent statistical issues this is the most accurate characterization of the value of factor timing, because it takes into account the information

²⁴ The numbers we report correspond to $\gamma = 1/2$.

behind these strategies in the portfolio evaluation. However, this approach does lean more heavily on our estimated model: because expected utility is an ex-ante concept, one needs to take a stand on the distribution to evaluate it. For this reason we can only report full-sample estimates of this quantity. For ease of comparison we report expected utility scaled as in the formulas from the beginning of this section. The gains to factor timing are large: they yield an almost twofold increase in expected utility from 1.66 to 2.96. Most of that increase, 1.26, comes from pure anomaly timing alone. In contrast, adding market timing to factor investing only slightly increases expected utility by 0.03; removing it from factor timing decreases expected utility by 0.04.

As we have discussed in Section 1.2, another interpretation of this utility calculation is in terms of volatility of the implied SDF. Therefore, these numbers suggest large differences in SDF behavior relative to estimates that ignore the evidence of factor predictability. We examine these changes and their economic implications in the next section. Before doing so, we briefly discuss the role of the portfolio rebalancing frequency.

3.2 How fast does timing need to be?

Our factor timing portfolios change weights on the anomalies each month. What happens if we trade more slowly? In Internet Appendix Table A.4, we report the performance measures for versions of the “pure anomaly timing” strategy that rebalances at lower frequencies. Performance does not deteriorate meaningfully. The unconditional Sharpe ratio actually increases from 0.71 to 0.79 with annual rebalancing. Expected utility declines from 1.26 to 0.81, still a substantial value. We discuss the other results and their construction at more length in Internet Appendix Section II. These results have two implications. First, the potentially less economically meaningful high-frequency variations in our predictors are not the main force behind our predictability results. Second, factor timing strategies might be implementable by actual investors; direct measures of transaction costs would be necessary to make this a firm conclusion.

4. Properties of the SDF

We now use the equivalence between optimal portfolio and SDF to study how including the possibility for factor timing affects the properties of an estimated SDF. This is of particular interest for the construction of economic models: because the SDF encodes asset prices, it can often directly be related to fundamentals of the economy in specific models. For example, the SDF is equal to the marginal utility of consumption of the representative agent in unconstrained economies.

4.1 Volatility

A first property of the SDF that has been the subject of attention by economists is its variance. Hansen and Jagannathan (1991) first show that the variance

Table 7
Variance of the SDF

	Factor investing	Market timing	Factor timing
$E[\text{var}_t(m_{t+1})]$	1.67	1.71	2.96
$\text{std}[\text{var}_t(m_{t+1})]$	—	0.29	2.17

We report the average conditional variance of the SDF and its standard deviation constructed under various sets of assumptions. “Factor timing” is our full estimate, which takes into account variation in the means of the PCs and the market. “Factor investing” imposes the assumption of no factor timing: conditional means are replaced by their unconditional counterpart. “Market timing” only allows for variation in the mean of the market return.

of the SDF is the largest squared Sharpe ratio attainable in the economy under complete markets; an upper bound in incomplete markets. Therefore, the variance of an SDF backed out from asset prices tells us how volatile the SDF has to be in an economic model to possibly account for this evidence. Hansen and Jagannathan (1991) document that the market return alone implies that the SDF is much more volatile than implied by reasonable calibrations of the CRRA model. Subsequent research has proposed models resolving this puzzle.

Naturally, using more conditioning information increases the investment opportunity set, and therefore the variance of the SDF. The results of Table 7 tell us by how much. We report the average conditional variance of the SDF constructed under various sets of assumptions. Remember that the SDF is given by: $m_{t+1} = 1 - \omega'_t(Z_{t+1} - \mathbb{E}_t[Z_{t+1}])$, where ω_t are the weights in the optimal portfolio. The conditional variance of the SDF is therefore:

$$\text{var}_t(m_{t+1}) = \omega'_t \Sigma_Z \omega_t, \quad (16)$$

which we then average over time. “Factor timing” is our full estimate of the SDF, which takes into account variation in the means of the factors, with ω_t given by Equation (11). “Factor investing” imposes the assumption of no factor timing: conditional means are replaced by their unconditional counterpart, Equation (12). Finally, “market timing” only allows for variation in the mean of the market return, Equation (13). These three names coincide with the portfolios of Section 3, because portfolio weights and SDF exposures are equal. However, there is a fundamental difference in interpretation. Within the same economy, one can evaluate separately the performance of various strategies. However, there is only one SDF that determines prices, so these various versions are just different estimators of this quantity. They cannot be all correct, and our “factor timing” specification is designed to be the best estimator, while the others force some misspecification in the construction. We find that accounting for the possibility to time factors substantially increases the estimated variance of the SDF. Adding this possibility in addition to timing the market and engaging in static factor investing yields estimates almost twice as large. The magnitude of the SDF variance, 2.96, is not only sizable compared to estimates ignoring factor timing but also substantially larger than in standard models. Bansal and Yaron (2004) report an annualized variance of the SDF of

0.85, whereas Campbell and Cochrane (1999) obtain a variance that roughly fluctuates between 0 and 1.2.

4.2 Heteroscedasticity

Next, we ask how much does the variance of the SDF change over time. Changes in variance of the SDF can capture how investors' attitude towards risk changes over time: when marginal utility is more volatile, investors ask for larger compensations for bearing risk. The observation that the SDF is heteroscedastic has long been known. Closely related to finding of predictability of market returns in Shiller (1981), many studies document that the Sharpe ratio of the market portfolio fluctuates. Motivated by this observation, a number of models have been developed to capture these changes. In the habit model of Campbell and Cochrane (1999), low realizations of consumption growth make households more risk averse, and renders the SDF more volatile. In the long-run risk model of Bansal and Yaron (2004) periods of high uncertainty about long-run consumption growth yield larger fluctuations in marginal utility.

Because the loadings of our estimated SDF on the components change over time, the SDF is heteroscedastic even though we assume the component returns themselves are not. Said otherwise, the maximum Sharpe ratio changes over time because prices of risk are changing. With our model estimates, we can compute not only the average variance of the SDF but also its conditional variance at each point in time using Equation (16). The solid-blue line in Figure 2 represents the result. The variance of our estimated SDF substantially varies over time: it fluctuates between low levels close to 0.8 and values as high as 12. The evidence of factor timing is the main driving force behind this result. As a comparison, we report estimates for an SDF estimated under the assumption of constant factor expected returns, but time variation in market risk premium. We depart from our baseline "market timing" specification and allow many variables other than the book-to-market ratio to forecast market returns.²⁵ Doing so yields an upper bound on what market return predictions using any combination of these variables imply in terms of SDF heteroscedasticity.²⁶ Despite this flexibility, the corresponding SDF variance is much less volatile than our estimate, with a volatility of only 0.29 relative to 2.17. In Internet Appendix Figure A.4, we also report the volatility of our estimated SDF if one imposes constant expected returns on the market and we find no meaningful difference relative to our estimate including factor timing. This observation further corroborates the importance of factor timing as a driver of the time variation in SDF variance.

²⁵ We include aggregate dividend/price, 10-year smoothed earnings/price, realized volatility, term premium, corporate spread, *cay* (Lettau and Ludvigson 2001), GDP growth, and sentiment (Baker and Wurgler 2006).

²⁶ Internet Appendix Figure A.4 reports the variance of our baseline "market timing" SDF estimate, which is indeed much less heteroscedastic than this specification.

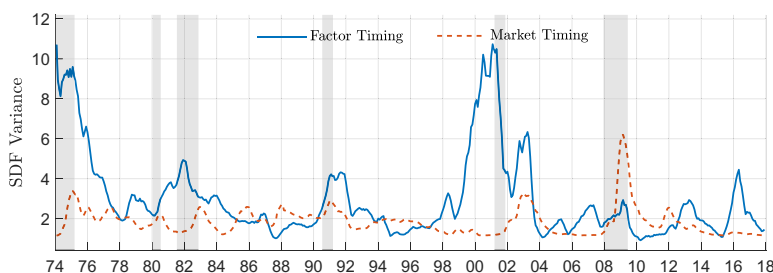


Figure 2
Conditional variance of SDFs

This figure plots the conditional variance of the SDF, constructed in two ways. The solid blue line represents the “factor timing” construction, which allows for variation in the means of the PCs and the market. The red dashed line represents a “market timing” estimate that ignores predictability of the anomaly factors. The aggregate market is forecast using aggregate dividend/price, 10-year smoothed earnings/price, realized volatility, term premium, corporate spread, *cay* (Lettau and Ludvigson 2001), gross domestic product (GDP) growth, and sentiment (Baker and Wurgler 2006).

The large variations over time we find in SDF variance are at odds with standard theories behind these changes. The variance of the SDF in Campbell and Cochrane (1999) has a standard deviation lower than 0.5. Models focusing on intermediary leverage such as He and Krishnamurthy (2013) generate large spikes in this variance when financial institutions are constrained because they become unwilling to absorb extra risks. However, these theories have a limited appeal in explaining the evidence from factor timing: the large spikes we observe in our estimates do not coincide with periods of intense stress in the financial sector.

4.3 Relation with economic conditions

Economic theories typically focus on specific drivers of variations in the maximum Sharpe ratio. To understand which properties these drivers must exhibit in order to rationalize our findings, we study how the variance of the SDF we estimate relates to measures of economic conditions. We standardized these variables to make estimates comparable.

A first observation is that the variance of the SDF exhibits a moderate degree of persistence. Its yearly autocorrelation is of 0.51. As we noticed when studying performance measures, this implies that following our signals even on a yearly basis provides similar results. Such a result is encouraging for macroeconomic models to explain this variation. However, 0.51 is also a much lower value than what is implied by theories focusing on slow, long-term changes in economic conditions. In line with these short-run patterns, the variance of the SDF is, *on average*, related to the state of the business cycle. It averages 4.9 during recessions while only being 2.7 during expansions. However, as one can notice by examining Figure 2, this relation is not systematic. In particular, the depth of the recession does not appear strongly related to the size of the spike in expected returns. This is in part because the market component in our “factor timing”

Table 8
SDF variance and macroeconomics variables

	Factor timing		Market timing	
<i>D/P</i>	−0.01	(0.03)	0.22	(2.86)
<i>GDP</i> growth	−0.37	(1.60)	−0.34	(4.93)
Market volatility	0.44	(2.48)	0.30	(5.37)
Sentiment	−0.15	(0.60)	−0.25	(3.35)
Common idio. volatility	0.49	(2.19)	−0.07	(0.97)
<i>cay</i>	−0.42	(1.79)	0.23	(3.00)
Term premium	−0.53	(2.36)	0.24	(3.35)
Inflation	0.75	(3.26)	0.01	(0.08)

We report univariate regression coefficients and absolute *t*-statistics from regressions of estimated SDF variance on various macroeconomic variables. Factor timing uses the SDF variance shown in Figure 2. Market timing uses SDF variance assuming the anomaly returns are not predictable and sets the conditional expected market return to the fitted value from a multivariate regression of market returns on the aggregate dividend/price ratio, realized market volatility, term spread, corporate bond yield spread, *cay* (Lettau and Ludvigson 2001), GDP growth, and aggregate sentiment (Baker and Wurgler 2006). Common idiosyncratic volatility is orthogonalized to market volatility (Herskovic et al. 2016).

SDF does not pick up large variations, unlike the one we use to construct the “market timing” SDF of Figure 2. But this also comes from the weak relation between the intensity of increases in anomaly expected returns with the overall intensity of broad macroeconomic events. For example, these two effects play a role in the low SDF volatility during the 2008 financial crisis relative to the tech boom and bust.

Table 8 relates the variance of the SDF to a variety of measures of economic conditions. For each measure, we report the coefficient and *t*-statistics of a univariate regression of the SDF variance on the variable. The first column uses our estimated SDF, which accounts for the factor timing evidence. The second column uses the estimate of SDF, which assumes constant expected returns on the factors, and forecasts market return using a variety of predictors, as in Figure 2. For this second column, because the market forecast is constructed using these specific conditions, obtaining significant relations is a somewhat mechanical process. However, the magnitude of the various coefficients still provides a useful benchmark. In contrast, for the first column, we did not use any of these variables are predictors.

We find that the dividend-price ratio of the market, a slow-moving measure of market conditions, does not predict the SDF variance. This is consistent with our discussion on the relatively low persistence of our estimated SDF variance: while *D/P* contains useful information for the market premium, it does not reveal much about overall fluctuations in Sharpe ratios, which move faster. More transitory measures of business cycle conditions do better: the variance of the SDF is larger following years of lower GDP growth and in periods of high market volatility or low economic sentiment, with loadings similar to those of the pure market timing SDF. Importantly, this similarity in loadings is not mechanically driven by the fact that our estimated SDF times the market: we obtain similar coefficients, −0.33 and 0.45, when setting the expected market return to a constant. Relatedly, the quantity of idiosyncratic

risk also becomes important for the variance of the SDF once one accounts for the evidence of factor timing. We find that the common idiosyncratic volatility measure of Herskovic et al. (2016), orthogonalized to market volatility, is strongly positively related to the variance of the factor timing SDF, but not of the market timing SDF.²⁷

Interestingly, some measures of economic conditions that predict the market positively, *cay* and the term premium, forecast the SDF variance negatively. Along these dimensions, risk compensations tend to be low for anomalies when they are large for the market. Finally, last year's inflation rate appears to strongly positively correlate with the SDF variance, while it holds no relation to the overall market return. Upon inspection of the time series (see Internet Appendix Figure A.5), we recognize that this relation appears to be driven by the bouts of high inflation in the early part of the sample, and the high inflation during the Internet boom.

Naturally, one should be wary of drawing any causal conclusion from these relations. That being said, economic theories associating variations in the variance of the SDF, or the overall appetite for risk, to these macroeconomic quantities, should confront this evidence to assess if they are consistent with the cross-section of expected returns.

4.4 Combining factor timing and volatility timing

So far, we have focused our efforts on using information from the predictability of factor returns to discipline the properties of the SDF. Movements in the volatility of factor returns could also play a potentially important role, as suggested by, for example, Moreira and Muir (2017).²⁸ In Internet Appendix Section II, we measure $\Sigma_{Z,t}$ and revisit the properties of the mean and standard deviation of the SDF variance. Including a time-varying $\Sigma_{Z,t}$ pushes the average SDF variance up from 2.97 to 3.54, about one-third of the increase from 1.66 provided by variation in risk premiums. In short, both changes in expected returns and variances contribute to the variance of the SDF, but the largest part comes from expected returns. The standard deviation of the squared Sharpe ratio goes from 2.17 to 2.06 when adding volatility timing to factor timing. This small change indicates that including information from changing volatility of returns only has a small effect on the cyclical behavior of the variance of the SDF. These results imply that the conclusions we have reached so far using means only are robust to including volatility timing.

4.5 What are the priced risks?

In addition to specifying how the size of risk compensations evolve over time, economic models typically also specify which sources of risk receive this

²⁷ We thank Bernard Herskovic, who graciously shared his data with us.

²⁸ Moreira and Muir (2017) document that variations in volatility dominate variations in expected returns in creating changes in the squared Sharpe ratio. They also show that volatility timing is useful for factor strategies.

compensation. For example, the basic tenant of the consumption-CAPM is that the SDF is proportional to aggregate consumption growth. Richer theories include other economic shocks, which are priced risks, such as long-run shocks to consumption growth, changes in disaster probability, or changes in the health of the financial sector. Going from our estimates to answering which sources of risks are priced is more challenging. When the SDF combines multiple, potentially correlated, sources of risk with time-varying loadings, it is generally not possible to characterize it without a complete structural specification. However, without focusing on a specific model, we can produce a number of statistics that are able to guide the design of future models.

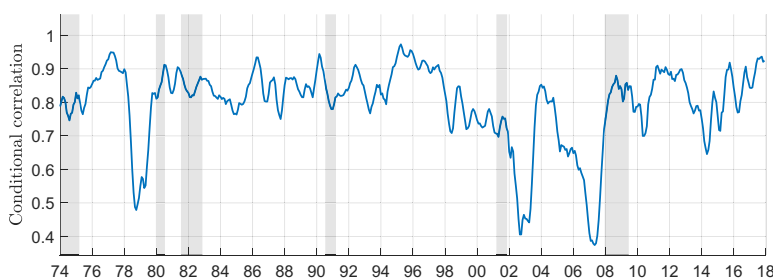
A first question we answer is whether using a model that targets an SDF that only combines static factor strategies is at least focusing on the right sources of risks. So far, we have showed that such an estimate grossly underestimates the variance and heteroscedasticity of the SDF. However, it might be feasible to easily “patch” univariate dynamics onto the estimate that fits the unconditional properties of returns. Concretely, start from the misspecified static specification of the SDF: $m_{t+1} = 1 - b' \varepsilon_{t+1}$. Is it enough to enrich the model with a single state variable x_t to obtain a SDF $m_{t+1} = 1 - x_t b' \varepsilon_{t+1}$ in order to capture the evidence of factor timing benefits? An economic motivation behind such x_t could be, for example, a shifter of the risk appetite of the economy, keeping constant the nature of economic risks investors worry about. For example, the habit model behaves conditionally a lot like the regular consumption-CAPM, but the level of habit affects risk aversion. In the intermediary asset pricing literature such as He and Krishnamurthy (2013) changes in intermediary wealth (or leverage) drive expected returns in all assets in which they are the marginal investor. Less formally, Stambaugh, Yu, and Yuan (2012) articulate a view where investor sentiment coordinates expected returns across anomalies.

This scaled SDF would be perfectly conditionally correlated with the baseline SDF, but would just add a time-varying loading on it. To assess whether this approach can work, we compute the conditional correlation of the misspecified SDF, which is estimated under the assumption of constant expected returns (“factor investing”) with our complete estimate of the SDF (“factor timing”). This correlation is given by

$$\text{corr}_t(m_{\text{F.T.},t+1}, m_{\text{F.I.},t+1}) = \frac{\omega'_{\text{F.T.},t} \Sigma_Z \omega_{\text{F.I.}}}{\sqrt{(\omega'_{\text{F.T.},t} \Sigma_Z \omega_{\text{F.T.},t})(\omega'_{\text{F.I.}} \Sigma_Z \omega_{\text{F.I.}})}}, \quad (17)$$

where the weights are defined in Equations (11) and (12). Figure 3 reports this conditional correlation. On average the two SDF estimates are quite correlated, with values that fluctuate around 0.8. However, the correlation exhibits strong time-series variation, with dips to values as low as 0.4. Both the observations of a mean meaningfully below 1 and these changes indicate that our SDF is not just a rescaling of the naive estimate ignoring factor timing evidence.

This first result suggests that one needs to include multiple sources of time-varying loading on shocks to fit the factor timing evidence. What should these

**Figure 3**

Conditional correlation of the estimated SDF and the misspecified SDF under the assumption of no factor timing benefits

This figure plots the conditional correlation of our estimated SDF and the misspecified SDF, which sets conditional factor means to their sample averages. Reported values are 6-month averages.

loadings look like? We can shed light on this question by considering how the covariance of our estimated SDF covaries with some specific anomalies. While there is no a priori reason that anomaly portfolio returns coincide with economic shocks, many theories offer a clear mapping between structural shocks and characteristic-sorted portfolio returns. For example, Hong and Stein (1999) develop a model of featuring both underreaction and overreaction to news, generating value and momentum type effects. In Papanikolaou (2011), the relative returns of value and growth stocks reveal investment-specific technological shocks. Alti and Titman (2019) study how investor overconfidence and aggregate disruption shocks lead to time-varying expected returns on value, profitability, and asset growth anomaly strategies. Berk, Green, and Naik (1999) study how the dynamics of firm investment lead to the value and momentum anomalies. With our estimates, it is straightforward to compute the conditional covariance of a specific factor return with the SDF: it is their conditional expected returns. Figure 4 reports the conditional expected return on four standardized factor returns: size, value, momentum and ROA.²⁹ Several interesting patterns emerge. All four strategies exhibit substantial variation in their correlation to the SDF, with frequent sign changes. Second, the pattern of these correlation differs across assets: size and value often have large correlations with the SDF during recessions, while momentum tends to relate negatively with the SDF during these episodes.³⁰ These patterns are not necessarily only driven by recessions: a lot of the cyclical movement in the correlation of the ROA factor with the SDF occurs outside of business cycle. Of course, these are just a few of the many potential portfolio sorts

²⁹ Because the factor strategies are correlated, the covariance of a specific factor with the SDF differs from the loading (or multivariate regression coefficient) of the SDF on this factor. We report these loadings in Internet Appendix Figure A.2.

³⁰ Interestingly, this last observation is consistent with the findings of Daniel and Moskowitz (2016), who document forecastable momentum crashes, even though we use completely different variables as predictors.

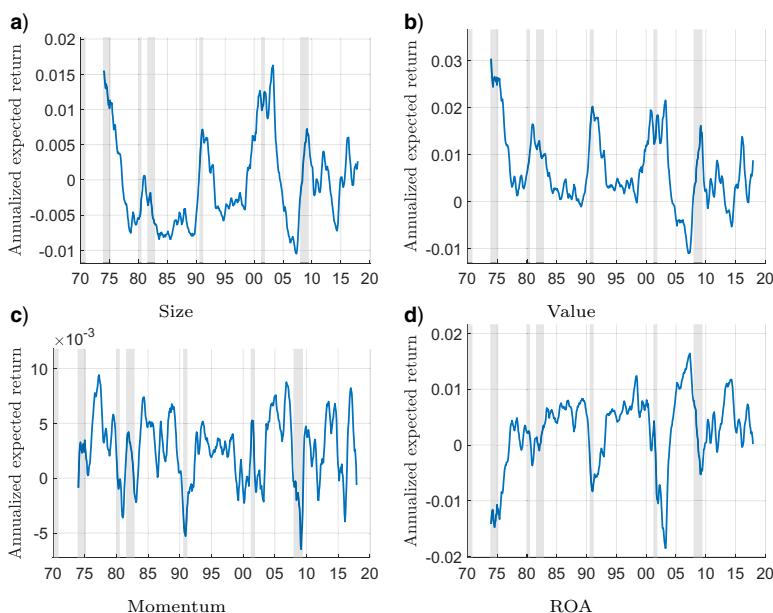


Figure 4
Anomaly expected returns

The plot shows the time-series of conditional expected returns on four anomaly strategies: size, value, momentum, and return on assets.

one can focus on. Internet Appendix Figure A.3 reports the average return and the standard deviation of conditional expected returns for each of the fifty portfolios. The figure shows a lot of cross-sectional variation in degree of predictability across anomalies. Interestingly, predictability and average performance are not related: the cross-sectional correlation between the mean and standard deviation numbers is close to zero. This result further highlights that there is not a single variable scaling all risk premiums up and down.

The broader point is that one can, with our estimates in hand, ask whether their theory generates a time-series pattern of loadings of the SDF on a specific shock or factor that is consistent with the evidence from the entire cross-section.

5. Concluding Remarks

In this paper, we study factor timing, which combines the ideas of long-short factor investing and market timing. Measuring the benefits to such strategies is relevant given their recent popularity, but more importantly, because they affect inference about the stochastic discount factor. However, taking a holistic approach to this question is challenging: the fact that one could time many potential factors increases the scope for finding spurious results. We use the idea of no near-arbitrage to overcome this issue. We show this principle disciplines

the estimation of the dynamics of expected returns. Thus guided, we obtain robust estimates of factor predictability. We find that factor timing is very valuable, generating superior performance relative to market timing and factor investing alone.

This conclusion has important consequences for the SDF. When we include our findings in the estimation of the SDF, we uncover a behavior that strongly differs from misspecified SDF estimates ignoring factor predictability. The variance of the implied SDF is much larger and more variable over time. These observations pose strong challenges for existing economic models if they aim to match the cross-section of returns, because the models understate these quantities. But the difficulty is not only about explaining a more volatile and heteroscedastic SDF. We find that the variance of the SDF exhibits different cyclical pattern than standard estimates, suggesting that some of the previously established drivers of variation in risk premiums are less important when looking at the cross-section. In addition, the dynamics of risk premiums are heterogenous across different factors.

In short, our results suggest that it is at least as important for economic theories to understand the dynamic properties of the cross-section as its average properties. We anticipate these facts will be helpful in guiding future research.

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